Extraction or Displacement? Labor Institutions and Surplus Distribution in India's Textile Value Chains

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by

Ashwin Sreekumar

CHRIST (Deemed to be University, Bangalore)

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Supervisor:

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Abstract

This dissertation investigates the mechanisms of surplus distribution between formal and informal sectors in India's textile and apparel manufacturing value chains. It addresses a central debate in development economics: whether the persistent linkage between formal and informal production reflects efficient technological complementarity (the "displacement" view) or is driven by unequal bargaining power enabling the extraction of rents (the "extraction" or "adverse incorporation" view). The ambiguity between these narratives—one driven by productivity and the other by power—has significant implications for labor and industrial policy.

The study develops a novel theoretical and empirical framework to distinguish between these competing explanations. Theoretically, it models a formal firm's decision to outsource tasks to informal suppliers using a nested Constant Elasticity of Substitution (CES) production function. The price of this outsourced work is determined through a Nash bargaining game, where the informal sector's bargaining power is explicitly modeled as a function of state-level labor law enforcement intensity. This maps the displacement view to the technological parameter (elasticity of substitution, ρ) and the extraction view to the bargaining power parameter (μ).

Empirically, the dissertation employs a two-stage methodology on a uniquely constructed district-industry panel dataset for 2023-24, integrating formal sector data from the Annual Survey of Industries (ASI) with informal sector data from the Annual Survey of Unorganized Sector Enterprises (ASUSE). Stage one uses a reduced-form analysis to establish that the pass-through of formal sector productivity gains to informal wages is significantly stronger in Indian states with stricter labor enforcement. Stage two employs a Simulated Method of Moments (SMM) estimator to structurally recover the deep parameters of the model— ρ and μ—using key moments of the data, such as the mean and variance of outsourcing intensity and the formal-informal wage gap.

This research makes three key contributions. First, it provides the first quantitative estimates of how labor institutions shape surplus distribution in India's formal-informal manufacturing linkages, offering a measure of the institutional "extraction rate." Second, methodologically, it demonstrates how to integrate political economy intuitions about power and institutions into a structural Industrial Organization framework, validating the 'extraction' hypothesis through revealed firm behavior rather than just wage differentials. Third, its findings inform critical policy debates on labor law reform in India, showing that regulatory "flexibility" may benefit formal capital at the direct expense of informal labor, with the welfare trade-offs hinging on the underlying technological relationship between the two sectors.

Keywords: Informal Sector, Formal-Informal Linkages, Bargaining Power, Labor Institutions, Value Chains, Structural Estimation, Indian Manufacturing, Textile Industry.

Chapter 1

Introduction

1.1 Problem Statement

The relationship between formal and informal sectors in developing economies has long been a central concern in development economics and public policy. India's dual economy presents a particularly compelling case study, with the informal sector employing over 80% of the non-agricultural workforce (International Labour Organization 2022). Economic development in dual economies has traditionally been theorized through a modernization lens, wherein the formal, modern sector inevitably absorbs or displaces the traditional, informal sector (Lewis 1954). On the other hand from 1990s a growing literature on formal-informal linkages rejects the dual economy perspectives and focus on the nature of linkages and their effect on the developmental prospects of the informal economy (Sethuraman 1992; Harriss 1990).

Existing literature lacks a quantifiable method to distinguish between these narratives. While studies show a link between formal and informal sectors, it remains unclear whether this link is driven by technological efficiency (displacement) or unequal bargaining power (extraction). This ambiguity hinders effective policy formulation: if displacement dominates, policy should focus on enhancing informal sector productivity; if extraction dominates, regulation must address power imbalances in supply chains. To resolve the debate between the Displacement View (driven by technological efficiency) and the Extraction View (driven by unequal power), this dissertation maps these narratives to structural parameters—specifically the elasticity of substitution (ρ) and bargaining power (μ).

1.2 Research Question

The dissertation asks:

1. To what extent does the pass-through of formal sector productivity to informal wages vary across states with different levels of labor enforcement?

2. Can the observed heterogeneity in wage responses be identified as a shift in bargaining power (μ) within vertical supply chains, as opposed to purely technological differences?

3. Given the estimated elasticity of substitution (ρ), how do shifts in labor enforcement redistribute welfare between formal capital, informal labor, and consumers (The Policy Counterfactual)?

1.3 Road map

A two-stage approach is adopted to study the informal-formal link in Indian manufacturing. Stage 1 establishes the empirical facts using a Reduced Form model, while Stage 2 uses Structural Estimation (SMM) to recover the unobserved bargaining power and productivity parameters. By explicitly modeling how state institutions—specifically labor enforcement—determine the distribution of value between formal capital and informal labor, this research provides the quantitative test of the "Adverse Incorporation" hypothesis in Indian manufacturing.

The structural parameters, enables us to run 'What-if' simulations. For example, what happens to the wage gap if we double enforcement? The Reduced Form only gives a local effect, but the Structural model allows us to see the equilibrium shift for the whole sector.

If full structural estimation proves infeasible within the dissertation timeline, the reduced-form analysis will stand as the primary empirical contribution, with the structural framework used for interpretation.

1.4 Expected Contributions

This dissertation makes three contributions:

1. **Empirical:** First quantitative evidence on how labor institutions shape surplus distribution in formal-informal value chains in India. Provides an estimate of the "extraction rate" and its institutional determinants.

2. **Methodological:** Demonstrates how to integrate political economy intuitions (power, institutions) into structural IO frameworks. The two-stage approach (reduced-form + structural) offers a template for other contexts.

3. **Policy:** Informs debates on labor law reform in India. Shows that "flexibility" (weaker enforcement) may benefit formal firms at informal workers' expense, with the magnitude depending on technological substitutability.

Chapter 2

Literature Review

Policy discussions regarding growth of developing countries often revolve around the role of informal economy. Informality remains a key concern of the Decent Work Agenda and the 2030 Agenda for Sustainable Development. The informality rate is an integral part of the indicators selected to measure progress towards the Sustainable Development Goals, as SDG indicator 8.3.1. It was speculated that with economic growth informal economies would diminish giving way to formalization. On the other hand formal-informal linkages have deepned. This dissertation contributes to understanding this puzzle by examining whether such linkages reflect efficient technological complementarities (the "displacement" view) or extractive power asymmetries (the "adverse incorporation" view).

This literature review is organized around four interconnected themes. First, the dissertation traces the theoretical models from dualist view of informality to structuralist perspectives that emphasize the functional role of informal labor in global value chains. Second, a review empirical evidence on formal-informal linkages through subcontracting and outsourcing, with particular attention to the Indian manufacturing context is carried out. Third, the literature review explores studies on labor institutions and their enforcement, which forms a crucial the theoretical foundation for identification strategy later. Finally, I discuss methodological approaches to estimating bargaining power and surplus distribution in vertically-linked production systems.

2.1 Theoretical Frameworks: From Dualism to Structural Heterogeneity

2.1.1 Dualism of Production

(Lewis 1954) differentiated the subsistence sector from the capitalist sector by articulating it as as that part of the economy which is not using reproducible capital. Compared to the capitliat sector in the subsistence sector output per head is lower because it is not fructified by capital Subsequent extensions made by Harris and Todaro (1970) and Rauch (1991) helped in the development of the classical view of informality. According to the dual model developing economies are characterized by a bifurcated structure: a modern, capital-intensive formal sector coexisting with a traditional, low-productivity informal sector.

(Lewis 1954) also posited that economic development proceeds through the gradual transfer of surplus labor from the traditional to the modern sector, with the process continuing until the labor surplus is exhausted and wages begin to rise across the economy. This perspective treats informality as a transitory phenomenon which will be replaced by the modern formal speher of economic activity with development. Subsequent refinements incorporated migration costs (Harris and Todaro, 1970), credit constraints (Evans and Jovanovic, 1989), and regulatory barriers (Rauch, 1991), but retained the core prediction: as institutional quality improves and capital deepens, informal firms either formalize or exit, unable to compete with the superior productivity of formal enterprises.

The displacement view finds empirical support in large productivity gaps between formal and informal firms. La Porta and Shleifer (2014) document that informal firms in developing countries exhibit productivity levels 30-70% below their formal counterparts, lack access to credit and modern technology, and rarely transition to formality. They argue that informal firms survive by evading taxes and regulations, not through genuine competitive advantage.

2.1.2 The Structuralist View of Production

A fundamentally different perspective emerges from structuralist political economy, which views informality not as a relic of underdevelopment but as a functional requirement of contemporary capitalism. Drawing on dependency theory and world-systems analysis, scholars such as Portes, Castells, and Benton (1989), Sassen (1996), and Meagher (2010) argue that informal labor is systematically incorporated into formal production networks on highly unequal terms—a process termed "adverse incorporation." Sanyal (2007) distinguishes between "need economy" activities (subsistence-level production for survival) and "formal economy" production that is integrated into capitalist accumulation. Crucially, he demonstrates how capital strategically deploys informal labor to extract surplus value while externalizing social reproduction costs. Rather than competing with formal firms, informal producers serve as subordinated suppliers, bearing volatility and regulatory risk while formal firms capture profits.

This perspective finds resonance in the global value chain (GVC) where scholars have shown that globalization post globalization the structure of global capitalism is not a uniform march toward formality. Milberg and Winkler (2013) by analyzing the relationship between offshoring and labor share in the U.S. and other OECD countries conclude that offshoring reduces the labor share, especially in countries with weaker labor market institutions. In addition, for developing economies participation in GVCs does not automatically lead to higher wages or better labor standards even when economy is upgrading, for example in the case of Mexico.

The GVCs that have emerged therefore present the case of powerful lead firms outsourcing to small, dependent suppliers in developing countries. In order to meet the demanding requirements (low cost, fast turnaround, flexibility) while bearing high risks and thin margins, these suppliers functionally depend on informal practices (Gereffi, Humphrey, and Sturgeon 2005). While Gereffi et al. (2005) provided the structural logic for why global capitalism fragments production, Barrientos, Gereffi, and Rossi (2011) reveal the consequences for labor. Diving deeper into how suppliers manage to maintain the required demand detail out how they employ regular workers to secure quality and consistency of production and irregular workers to cope with fluctuating orders and downward price/cost pressures. Their findings on the Mo-

roccan garment case show even though companies might take complex taxes and economically upgrade they rely on a periphery of irregular, informal workers (often young, female migrants) for low-skilled, time-sensitive tasks (packing, loading) under poor conditions (Barrientos, Gereffi, and Rossi 2011). Far from being a separate 'traditional' sector which was destined to phase out, informality has remained as a functional requirement and direct outcome of the competitive dynamics at the core of buyer-driven, globally fragmented production. The GVC framework thus reveals informality as constitutive of, not external to, modern capitalism."

2.1.3 Debates on Informality in India

The Indian context provides particularly rich terrain for examining formal-informal linkages due to its distinctive labor regulatory framework. India inherited a comprehensive system of protective labor legislation from the colonial and early post-independence periods, including the Industrial Disputes Act (1947), Factories Act (1948), and Contract Labour Act (1970). However, enforcement of these laws varies dramatically across states and over time, creating quasi-experimental variation in institutional environments. Besley and Burgess (2004) exploit amendments to the Industrial Disputes Act across Indian states between 1958-1992, coding them as "pro-worker," "pro-employer," or "neutral." They find that pro-worker amendments are associated with lower output, employment, investment, and productivity in registered manufacturing, with the effects concentrated in labor-intensive industries. Importantly, they find that restrictive labor regulations increase informal sector employment, suggesting substitution between formal and informal labor. Ahsan and Pagés (2009) extend this analysis using a continuous measure of labor regulation stringency. They confirm that stricter regulations reduce formal manufacturing growth but provide evidence of heterogeneous effects: small firms are disproportionately affected, consistent with fixed compliance costs. Bhattacharjea (2006) cautions that the Besley-Burgess coding scheme conflates amendments that affect workers' collective bargaining rights with those affecting firms' adjustment costs, potentially obscuring the mechanisms at work. More recent work has focused on enforcement rather than statutory regulations. Chaurey (2015) shows that increases in labor inspections reduce formal employment but do not affect output or productivity, suggesting that firms respond by shifting toward con-

tract labor or outsourcing. Adhvaryu, Chari, and Sharma (2013) find that firms located in areas with more frequent inspections are more likely to engage in "spurious compliance"—recording workers as contract laborers to evade permanent employment protections. Critically, this literature establishes that both regulation and enforcement matter, but it largely treats informality as a labor supply response (workers pushed into self-employment) rather than examining how formal firms strategically utilize informal suppliers to circumvent regulations. My dissertation addresses this gap by modeling the vertical linkage between formal firms' outsourcing decisions and informal producers' bargaining power.

2.2 Empirical Evidence on Formal-Informal Subcontracting Linkages

2.2.1 Global Evidence on Outsourcing and Labor Market Outcomes

A substantial empirical literature examines how firms' make-or-buy decisions affect labor market outcomes, though much of this work focuses on developed economies or international offshoring rather than domestic formal-informal linkages. Abraham and Taylor (1996) document the rise of outsourcing in U.S. manufacturing, finding that firms substitute purchased services for direct employment in response to demand volatility and union pressure. Autor (2003) shows that temporary help employment grew as firms sought flexibility, with wages for temporary workers 20% below comparable permanent employees. Dube and Kaplan (2010) find that outsourcing of food and janitorial services in U.S. hospitals reduces wages for affected workers by 4-7%, with the effect mediated through weaker bargaining power rather than productivity differences. Goldschmidt and Schmieder (2017) exploit German administrative data to show that domestic outsourcing accounts for approximately 10% of the rise in wage inequality between 1985-2008. They find that outsourced workers earn 10-15% less than in-house workers in identical occupations, even after controlling for match quality. Crucially, they demonstrate that wage losses persist for individual workers who transition from in-house to outsourced employment, ruling out pure selection effects. Studies of international offshoring

yield related insights. Harrison and McMillan (2011) find that manufacturing outsourcing to low-wage countries reduces domestic manufacturing wages and employment in the U.S., with effects concentrated among less-educated workers. Antràs, Fort, and Tintelnot (2017) develop a quantitative model of multinationals' sourcing decisions, showing that the elasticity of substitution between domestic and foreign inputs critically determines the distributional effects of globalization. 2.3.2 Subcontracting in Developing Country Manufacturing The literature on domestic subcontracting in developing countries is more limited but growing. Basu, Chau, and Kanbur (2010) develop a theoretical model where formal firms subcontract to informal firms that employ child labor, showing how global supply chains can inadvertently perpetuate exploitative practices. Fernandez-Stark, Frederick, and Gereffi (2011) document the "putting-out" system in the apparel industry across Asia and Latin America, where formal firms supply raw materials to home-based workers who perform labor-intensive tasks for piece-rate wages well below minimum wage. Marjit (2003) and Kar and Marjit (2009) model formal-informal linkages in the Indian context as a nested production structure where formal firms use informal services as intermediate inputs. They show theoretically that trade liberalization can increase demand for informal services if formal manufacturing expands, but they do not empirically test the mechanism or examine how bargaining power affects surplus distribution. Recent empirical work provides more direct evidence. Nataraj (2011) uses Indian plant-level data to show that formal manufacturing firms increase outsourcing following trade liberalization, particularly in states with restrictive labor regulations. She finds that informal manufacturing employment rises in these states, but does not examine wage effects. McCaig and Pavcnik (2018) study Vietnam's enterprise formalization reforms, finding that provinces with more rapid formalization exhibit slower wage growth in household enterprises, consistent with monopsony power by newly formal firms. Critical Gap: While this literature establishes that formal firms do outsource to informal producers and that this outsourcing is sensitive to labor regulations, it does not quantify the distribution of surplus between formal buyers and informal suppliers. Is the low profitability of informal firms explained by low productivity (as La Porta and Shleifer argue) or by extractive contract terms (as the adverse incorporation thesis suggests)? My dissertation directly addresses this question by structurally estimating the bargaining parameter μ.

2.3.3 The Indian Textile Value Chain: Institutional Context India's textile and garment sector provides an ideal setting for examining formal-informal linkages. The sector accounts for approximately 7% of manufacturing output and 12% of export earnings, with employment of over 45 million workers (Ministry of Textiles, 2023). Crucially, production is organized through extensive vertical disintegration: large formal mills and exporters subcontract labor-intensive tasks (embroidery, stitching, dyeing, finishing) to networks of small unincorporated workshops and home-based workers. Tewari (1999) provides rich ethnographic evidence on the organization of the Ludhiana knitwear cluster in Punjab, documenting how formal exporters coordinate production across hundreds of small job-workers without formal contracts. Kadiyali, Chintagunta, and Vilcassim (2000) analyze productivity spillovers in the Tiruppur garment cluster in Tamil Nadu, finding limited technology transfer from formal to informal units despite dense network ties. Soundararajan (2019) examines the effect of the Mahatma Gandhi National Rural Employment Guarantee Act (NREGA) on informal manufacturing wages. She finds that informal wages rise in districts with more intensive NREGA implementation, consistent with improved outside options for informal workers. However, she also finds evidence of declining informal employment, suggesting potential displacement effects. This finding underscores the importance of understanding the elasticity of substitution (ρ) between formal and informal inputs—a central parameter in my structural model. Singh (2023) provides the most directly relevant recent work, examining contract labor in Indian manufacturing using ASI data. He finds that formal firms increase use of contract labor in response to pro-worker amendments to labor laws, and that contract workers earn 20-30% less than permanent workers in observably similar jobs. However, Singh focuses on internal contract labor (workers employed on-site through contractors) rather than external outsourcing to separate enterprises. My study complements his work by analyzing Block F (job work done by others) rather than Block E (contract labor), isolating the inter-firm bargaining channel.

2.3 Labor Institutions, Enforcement, and Bargaining Power

2.3.1 Theoretical Frameworks: Institutions and Rent-Sharing

My identification strategy rests on the theoretical premise that labor institutions affect workers' bargaining power and hence their share of economic rents. This hypothesis has deep roots in labor economics, though formalization varies across approaches. In search-and-matching models (Mortensen and Pissarides, 1994; Pissarides, 2000), wages are determined by Nash bargaining between workers and firms over the surplus generated by the match. The worker's bargaining power parameter reflects institutional factors such as employment protection, unionization, and the generosity of unemployment insurance. Pissarides and Vallanti (2007) show that stricter employment protection increases workers' bargaining power by reducing the firm's threat point (the value of terminating the match). Rent-sharing models in the efficiency wage tradition (Shapiro and Stiglitz, 1984; Blanchflower, Oswald, and Sanfey, 1996) provide an alternative microfoundation. If firms pay above-market wages to elicit effort or reduce turnover, then institutions that raise the cost of dismissal or strengthen workers' outside options will tilt rent-sharing toward workers. Card, Devicienti, and Maida (2014) find that Italian firms share 15-30% of productivity gains with incumbent workers, with the pass-through higher in unionized plants. Most relevant to my setting is the literature on bargaining in vertical supply chains. Hart and Moore (1990) and Grossman and Hart (1986) develop property rights theories where asset ownership determines parties' threat points in ex-post bargaining. Antràs (2003) applies this framework to international trade, showing that multinational firms' integration decisions depend on the hold-up problem: if suppliers make relationship-specific investments, they risk expropriation if the buyer has all the bargaining power. Application to Informal Outsourcing: In the Indian textile context, informal workers make implicit investments (learning specific embroidery techniques, purchasing specialized equipment) that are more valuable to their current formal buyer than to alternative buyers. If labor enforcement is weak, informal workers' threat point is low—they cannot credibly walk away to formal employment at minimum wage. Thus, weak enforcement endogenizes low bargaining power (μ),

which I test empirically.

2.3.2 Empirical Evidence on Enforcement and Wage Determination

While the theoretical link between institutions and bargaining is well-established, empirical identification is challenging because institutional quality is endogenous to economic development. Three identification strategies have proven fruitful: 1. Cross-country comparisons with institutional instrumentation. Botero et al. (2004) code labor regulations for 85 countries and find that stricter regulations are associated with lower employment and larger informal sectors. However, these regressions suffer from omitted variable bias (institutional quality correlates with rule of law, education, etc.) and cannot isolate causal effects. 2. Within-country policy changes. Aghion, Algan, Cahuc, and Shleifer (2010) exploit French reforms to workplace representation, finding that regions with stronger unions exhibit higher pass-through of productivity to wages. Lamadon, Mogstad, and Setzler (2022) use matched employer-employee data from Sweden to estimate firm-specific rent-sharing parameters, finding substantial heterogeneity: workers in high-rent-sharing firms capture 30-40% of productivity gains, while those in low-rent-sharing firms capture less than 103. Spatial discontinuities in enforcement. Bhorat, Kanbur, and Stanwix (2017) exploit minimum wage variation across sectors in South Africa, finding compliance rates of 60-80% even in sectors with weak unionization, suggesting that enforcement matters. Almeida and Carneiro (2012) show that Brazilian firms located closer to labor ministry offices face more frequent inspections and exhibit higher compliance with minimum wage laws, with effects concentrated among less-educated workers. Application to Indian States: The Besley-Burgess (2004) framework treats state-level labor law amendments as quasi-exogenous shocks, though Bhattacharjea (2006) and others have questioned this assumption. More promisingly, variation in enforcement intensity—inspector-to-firm ratios, inspection frequencies, dispute resolution patterns—offers a plausibly exogenous source of variation, particularly in cross-sectional analysis where I control for state fixed effects. My dissertation extends this literature by testing whether enforcement affects not just wage levels but the elasticity of informal wages with respect to formal productivity—the signature of bargaining power.

2.4 Methodological Approaches: Estimating Bargaining Power and Technology

2.4.1 Reduced-Form Approaches: Pass-Through Regressions

The canonical empirical approach to identifying rent-sharing regresses worker wages on firm productivity or profitability (Van Reenen, 1996; Abowd, Kramarz, and Margolis, 1999). If labor markets were perfectly competitive, wages would depend only on workers' marginal product, which reflects individual human capital. Any correlation between firm-level productivity and individual wages (conditional on worker characteristics) indicates rent-sharing. Card, Heining, and Kline (2013) implement this approach with German administrative data, using the Abowd-Kramarz-Margolis (AKM) decomposition to separate worker and firm wage components. They find that 20% of wage variance is attributable to firm fixed effects, and that high-productivity firms pay higher wages even to observationally identical workers. Lamadon et al. (2022) extend this framework to allow for heterogeneous rent-sharing parameters, estimating that workers' bargaining power varies from 0.1 to 0.5 across firms. Limitations for the Informal Sector Context: This approach requires matched employer-employee panel data to control for worker composition through individual fixed effects. Indian informal sector data (ASUSE) provides only cross-sectional establishment-level observations. I therefore adapt the pass-through framework to district-industry aggregates, testing whether the interaction between formal productivity and state enforcement predicts informal wages—a difference-in-differences style specification that allows for time-invariant state heterogeneity.

2.4.2 Structural Estimation: Simulated Method of Moments

Reduced-form regressions establish conditional correlations but cannot separately identify technology (the substitution parameter ρ) from bargaining (the power parameter μ). Structural estimation addresses this by explicitly modeling firms' optimization problems and using equilibrium restrictions to identify parameters. The Simulated Method of Moments (SMM), developed by McFadden (1989) and Pakes and Pollard (1989), is particularly well-suited to

settings where the model's equilibrium cannot be solved in closed form. The estimator minimizes the distance between moments computed from actual data and moments generated by simulating the model at candidate parameter values. Lee and Wolpin (2006) provide a comprehensive guide, emphasizing the importance of selecting moments that provide unique variation for each parameter.

Applications to production and trade. De Loecker (2011) uses SMM to estimate productivity distributions and markups in Belgian manufacturing, exploiting trade liberalization as a quasi-experiment. Hsieh and Klenow (2009) structurally estimate misallocation in Chinese and Indian manufacturing by matching moments of the firm size distribution to a model with heterogeneous distortions. Antràs et al. (2017) use a nested-logit structure to estimate multinational firms' sourcing elasticities, finding that domestic and foreign inputs are moderately substitutable ($\rho \approx 2$–3). Novelty of my approach is that I extend the SMM framework to a setting with both technological complementarity (through the CES production function) and strategic bargaining (through the Nash price-setting equation). The key innovation is to use institutional variation (state-level enforcement) to separately identify technology from power. States with identical production technologies (ρ, γ) can exhibit different informal wage outcomes because they differ in $\mu(E_s)$. This institutional variation disciplines the bargaining parameter while the dispersion of outsourcing intensity (conditional on costs) disciplines the substitution parameter.

Structural estimation faces two generic identification challenges: parameter multiplicity (multiple parameter vectors can fit the same moments) and weak identification (moments provide little information about certain parameters). Parameter Multiplicity: When multiple parameters affect the same outcome, the model may be under-identified. For example, both low substitutability ($\rho \to -\infty$, making formal and informal inputs complements) and low bargaining power ($\mu \to 0$, depressing informal wages) can produce the same observed wage gap. I address this by exploiting the different implications for quantities versus prices:

Chapter 3

Theoretical Framework

3.1 Model Specification

The central empirical challenge of this dissertation is to distinguish between two competing explanations for low informal sector wages in India's textile value chains. The displacement view attributes wage gaps to technological backwardness: informal firms simply produce less value per worker due to obsolete equipment, managerial inefficiency, and limited scale. The extraction view attributes wage gaps to unequal bargaining power: informal firms may be reasonably productive, but formal buyers can exploit their monopsony position to appropriate most of the surplus. Observationally, both stories predict the same outcome—informal workers earn less than formal workers. To distinguish between them, I need a structural model that makes each mechanism mathematically explicit and derives testable implications.

This chapter develops such a model, with three core components: The Decision to Outsource, The Institutional Bargaining Game and How Enforcement Affects Bargaining Power. To disentangle efficiency from power, I develop a structural model of the "Make vs. Buy" decision in the Indian Textile Industry (NIC 13 & 14).

3.1.1 The Decision to Outsource

A representative Formal Firm (F) in district d, state s and industry ii i maximizes profit by choosing between producing tasks in-house (using regulated labor L_F) or outsourcing them (using informal job-work M_{inf}).

The production function is modeled as a nested **Constant Elasticity of Substitution (CES)** technology. For tractability, I impose Constant Returns to Scale (CRS), consistent with

the assumption of perfect competition in the product market:

$$Y_{F,dsi} = A_{F,dsi}\left[\gamma L_{F,dsi}^{\rho} + (1-\gamma)M_{\text{inf},dsi}^{\rho}\right]^{1/\rho} \tag{3.1}$$

where:

- Y_F = Output of the formal firm
- A_F = Formal firm Total Factor Productivity (Hicks-neutral technical change)
- $\gamma \in (0,1)$ = Distribution parameter (share of formal labor)
- $\rho < 1$ = Substitution parameter. The elasticity of substitution is $\sigma = \frac{1}{1-\rho}$.
- M_{inf} = Quantity of outsourced informal tasks

In textiles, tasks like embroidery, dyeing, or stitching can be moved between a factory floor and outside workshops. The parameter ρ captures this substitutability. When σ is high (close substitutes), formal firms can easily switch between in-house production and outsourcing in response to relative price changes. When σ is low (complements), the production structure is more rigid.

3.1.2 The Institutional Bargaining Game

Unlike formal labor, which faces statutory minimum wages, the price of outsourced tasks (p_{sub}) is determined through negotiation between formal buyers and informal suppliers. Following the generalized Nash bargaining framework, the equilibrium price is:

$$p_{\text{sub}} = (1-\mu_s)\cdot MC_{\text{inf}} + \mu_s \cdot MRP_F \tag{3.2}$$

where:

- MC_{inf} = Marginal cost of informal production
- MRP_F = Marginal revenue product of the task for the formal firm
- $\mu_s \in [0,1]$ = Informal sector's bargaining power in state s

When $\mu_s = 0$, formal firms capture all surplus (full monopsony); when $\mu_s = 1$, informal suppliers capture all surplus (competitive outcome).

- **Hypothesis:** $\mu'(E_s) > 0$. As enforcement rises, the informal sector captures a larger share of the value it creates.

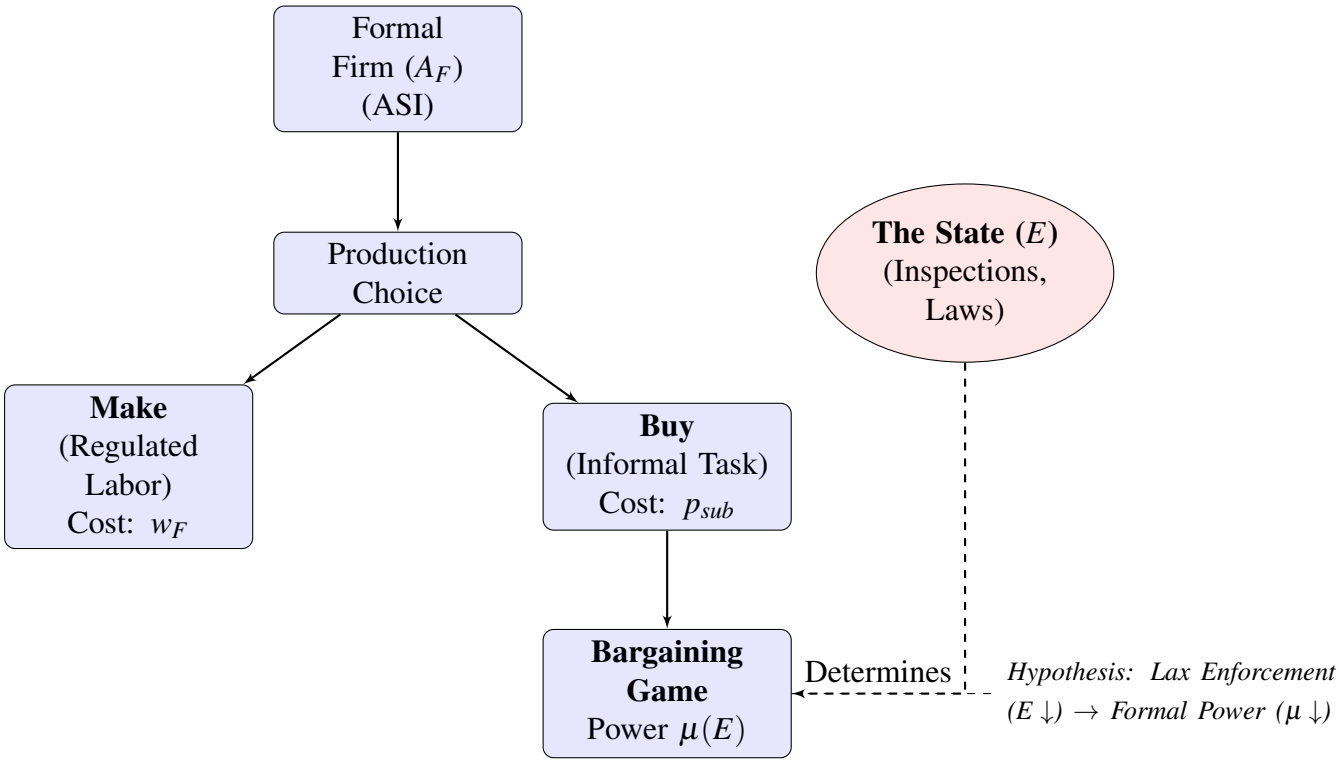

Figure 3.1: The Political Economy of Value Chains

3.1.3 How Enforcement Affects Bargaining Power

The key innovation of this model is to endogenize μ_s as a function of the state's enforcement environment.

In states with strict labor law enforcement, informal workers have credible access to formal sector employment at legally mandated wages. This raises their reservation wage—the minimum they will accept for informal work. Conversely, in states with weak enforcement, formal firms routinely violate minimum wage laws, so informal workers cannot credibly threaten to switch to formal employment. Their outside option is weaker, reducing their bargaining power.

Formally, let R_s denote the reservation wage in state s. This depends on:

- The statutory minimum wage $w_s^{\min}$
- The probability of enforcement $\pi(E_s)$, where $\pi'(E_s) > 0$

The effective outside option is:

$$R_s = \pi(E_s) \cdot w_s^{\min} + [1 - \pi(E_s)] \cdot w^{\text{outside}} \tag{3.3}$$

where w^{outside} is the fallback wage in the absence of enforcement (e.g., subsistence agriculture, casual labor).

In Nash bargaining, the informal surplus is $(p_{\text{sub}} - R_s)$ rather than $(p_{\text{sub}} - MC_{\text{inf}})$. Higher enforcement raises R_s, which increases the informal sector's bargaining power:

$$\mu_s = f(R_s(E_s)) \quad \text{where} \quad \frac{d\mu_s}{dE_s} = f'(R_s) \cdot \pi'(E_s) > 0 \tag{3.4}$$

For empirical implementation, I approximate this with a linear reduced form:

$$\mu_s = \beta_0 + \beta_1 E_s + \varepsilon_s \tag{3.5}$$

Hypothesis: $\beta_1 > 0$. States with stricter enforcement exhibit higher informal bargaining power, manifesting as greater "pass-through" of formal sector productivity gains to informal wages.

3.1.4 Equilibrium Specification

To make the model tractable for a Master's thesis, I adopt a **partial equilibrium** framework with the following assumptions:

1. **Product market:** Formal firms are price-takers in the output market (perfect competition). This allows me to treat output price P as exogenous.
2. **Firm entry:** The number and productivity distribution of formal firms in each district-industry is taken as given (no entry/exit dynamics). This is justified by the cross-sectional nature of the data.

3. **Informal supply:** Informal suppliers are atomistic. Each district-industry has a large number of informal firms with marginal cost MC_{inf} determined by local factor costs (wages, materials). Informal capacity is assumed elastic within the cross-section.

4. **Market clearing:** For a given district-industry, total demand for outsourced tasks by formal firms equals total supply from informal firms. This pins down the equilibrium quantity M_{inf}.

These assumptions allow me to focus on the *distribution* of surplus (via μ_s) conditional on productivity and costs, rather than modeling the full general equilibrium.

3.1.5 Interpretation: What μ_s Actually Captures

The estimated bargaining power parameter μ_s should be interpreted as a **reduced-form parameter** that captures the net effect of enforcement on surplus distribution. It implicitly embeds three distinct mechanisms:

1. **Direct outside option effect:** $\pi(E_s)$ raises the credibility of formal employment as an alternative for informal workers.

2. **Job availability effect:** Conditional on enforcement, the actual density of formal jobs in the local labor market determines whether the outside option is accessible. Even with perfect enforcement, if formal jobs are scarce (D_s is low), workers' bargaining position remains weak.

3. **Firm strategic response:** Enforcement may change firms' make-or-buy decisions, shifting the composition of tasks allocated to informal suppliers. If firms respond to stricter regulation by outsourcing more labor-intensive tasks, this could paradoxically increase informal employment while affecting bargaining dynamics.

Our empirical strategy disentangles mechanism (1) from (2) through the $E_s \times D_s$ interaction in the reduced-form analysis (Equation 4.6), and tests for (3) through the outsourcing

intensity regression (Equation 4.7). The structural parameter μ_s thus represents the **equilibrium bargaining power** after accounting for these combined forces, making it the appropriate reduced-form parameter for counterfactual policy analysis.

Chapter 4

Methodology and Data Strategy

4.1 Data Strategy

I construct a **District-Industry Panel** by linking two major datasets for the year 2023–24. I chose the textile manufacturing sector as "outsourcing embroidery" for example is a direct substitute for "hiring an embroidery worker." In Auto-parts, it is a complement. Focusing on Textiles also justifies the CES function. Informal textile workshops are structurally similar. Comparing a garment workshop to a steel smelter (in a full manufacturing model) adds too much noise.

4.1.1 Data Sources

1. **Formal Sector (ASI 2023–24):** Singh (2023) focuses on Block E (internal contract labor), this study focuses on Block F (Job Work). This distinction is critical as it isolates 'external extraction'—where the formal firm outsources the task entirely to avoid overhead—rather than just hiring flexible labor on-site.

    - *Outsourcing Intensity:* From **Block F: "Cost of materials consumed for job work done by others"**. This captures formal firms' expenditure on subcontracting.
    - *Productivity:* Total Factor Productivity (A_F) calculated using the value-added method with capital and labor inputs. *(See Appendix A.2 for full construction details.)*
    - *Formal wages:* From Block G (emoluments paid).

2. **Informal Sector (ASUSE 2023–24):**

- *Informal wages:* **Average wage per worker** in informal textile enterprises, calculated from Block 5 (operating expenses on hired workers) divided by number of hired workers. This is a more direct proxy for p_{sub} than GVA per worker, as it reflects the labor cost component that formal firms indirectly pay through subcontracting. *(See Appendix A.2 for measurement and deflation details.)*
- *Productivity:* Output per worker as a control variable.
- *Subcontracting flag:* ASUSE questionnaire asks whether the enterprise does job work for others (yes/no). I restrict the sample to informal firms reporting subcontracting activity.

3. **Institutional Data:**

Besley and Burgess's strategy of using State-Level Labor Amendments (or inspection rates) is used here as an exogenous shock (E). Instead of just checking if E affects Employment (L), you check if E modifies the relationship between Formal Productivity A_f and Informal Wages W_{inf}.

- *Enforcement (E_s):* State-level measures from the Ministry of Labour & Employment:
  - Inspector density (inspectors per 1000 factories)
  - Number of inspections conducted per registered factory
  - Proportion of labor disputes resolved in favor of workers
- *Alternative measure:* The Besley-Burgess (2004) Labor Regulation Index, updated for recent reforms.
- *Control variables:* State GDP per capita, literacy rate, urbanization rate, ruling party ideology (pro-labor versus pro-business).

4.1.2 Sample Selection and Aggregation

A critical methodological challenge is distinguishing between "Contract Labor" (informal workers on-site) and "Outsourcing" (production off-site). I utilize the structural distinction

in the ASI questionnaire:

- **On-Site Labor:** Workers employed through contractors are reported in Block E and are treated as part of the formal firm's labor force (L_F).
- **Off-Site Outsourcing:** Expenses on "Job work done by others" are reported in Block F. I define this as the outsourcing intensity (Q_{out}).

Assumption here is that "Job Work" expenses primarily capture flows to the informal/unincorporated sector. This is supported by the industrial organization of the Indian textile sector, where the "putting-out" system involves formal firms supplying raw materials to small, unorganized units for labor-intensive processing (embroidery, finishing) rather than B2B transactions between large formal entities (typically recorded as material purchases in Block H). To align the empirical analysis with this assumption, the informal sector sample (ASUSE) is explicitly restricted to enterprises reporting "receipts from job work" as a primary activity

1. **Industry:** Restricted to **Textiles (NIC 13) and Wearing Apparel (NIC 14)**. This ensures task homogeneity (stitching, embroidery, dyeing are comparable across regions) and adequate sample size in ASUSE.
2. **Geography:** District $\times$ State $\times$ NIC-2-digit level. I aggregate firm-level data to the district-industry cell to avoid thin-sample issues in ASUSE while preserving within-state variation.
3. **Minimum sample size:** Exclude cells with fewer than 10 observations in either ASI or ASUSE to ensure reliable moment calculation.

4.2 Empirical Strategy

To understand the dynamics of extraction and displacement in the textile industry, this study faces a fundamental challenge: **Power is invisible.**

While we can observe data points in the Annual Survey of Industries (ASI)—such as the wages paid, the number of workers hired, and the amount of outsourcing—we cannot directly observe the *bargaining power* of the formal firm or the *strategic intent* behind their

decisions. Standard statistical methods (like regression analysis) can tell us *if* enforcement is correlated with wages, but they struggle to explain why that relationship exists or *how* the surplus is actually split.

To solve this, this dissertation employs a **Structural Modeling** approach, specifically the **Simulated Method of Moments (SMM)**. For the non-specialist, this method can be understood as a process of "economic reverse engineering."

Imagine we are observing a car driving on a highway. We can see its speed (Data), but we cannot see how hard the driver is pressing the gas pedal (The Parameter). However, if we know the laws of physics and the engine's specifications (The Theory), we can calculate exactly how much pressure *must* be on the pedal to produce that specific speed.

In this study:

* **The "Speed" (Data):** We observe the real-world ratios of outsourcing, formal wages, and informal wages in the ASI and ASUSE datasets. * **The "Laws of Physics" (Theory):** We use **Nash Bargaining Theory**, which describes how two parties split a profit based on their relative strength. * **The "Gas Pedal" (Unknown Parameter):** We are trying to measure (Bargaining Power)—the invisible leverage the formal firm holds over the informal supplier.

The methodology proceeds in three intuitive steps:

1. Constructing the "Digital Twin": We build a mathematical simulation of a typical textile district. In this simulation, "virtual" firms make decisions about hiring and outsourcing based on the costs and labor laws we program into them. 2. The Guessing Game (Simulation): Initially, the computer does not know the true bargaining power of firms. It makes a guess (e.g., "Assume formal firms have 50% power"). It runs the simulation based on this guess and produces "simulated data" (e.g., simulated wage levels). 3. Matching Reality (Estimation): The algorithm compares its *simulated data* against the *real data* from Indian districts. * If the simulated wages are too high compared to reality, the computer realizes its guess was wrong (the firms must have more power). * It adjusts the "power dial" and re-runs the simulation. * It repeats this process thousands of times until the simulated world matches the real world almost perfectly.

By the end of this process, we obtain a mathematically estimated value for **Bargaining Power ()** that is consistent with the real-world data.

This allows us to move beyond simple correlations. Instead of just saying "Enforcement changes wages," we can quantify **how much of the economic surplus is being extracted** by formal firms and determining whether labor laws successfully force them to share that pie, or if they simply abandon the informal sector entirely. This distinction is the core of the "Extraction vs. Displacement" debate.

4.2.1 Stage 1: Reduced-Form Analysis of Institutional Wage Pass-Through

Conceptual Overview: A Two-Stage Approach

This dissertation employs a two-stage empirical strategy to distinguish between technological displacement and power-based extraction in formal-informal linkages:

1. **Stage 1 (Reduced-Form Analysis):** Establishes conditional correlations between enforcement, productivity, and wages through multiple regression specifications. This stage tests specific theoretical mechanisms:
   - Does enforcement affect wage pass-through? (Baseline: Eq. 4.1)
   - Does enforcement require job availability to be effective? (Job Availability: Eq. 4.6)
   - Do firms strategically outsource more under strict enforcement? (Strategic Substitution: Eq. 4.7)
2. **Stage 2 (Structural Estimation):** Uses the Simulated Method of Moments (SMM) to recover "deep parameters" (ρ, γ, β_0, β_1) that are invariant to policy changes. These parameters allow for counterfactual policy simulations that address the Lucas Critique.

The reduced-form analysis serves two purposes: (1) providing empirical evidence for the bargaining hypothesis, and (2) generating "moments" (statistical patterns) that discipline the structural model. This integrated approach allows us to move from correlation to causation while acknowledging the limitations of observational data.

This aggregate approach presents two identification challenges:

1. **Composition bias:** High-enforcement states may attract higher-skilled informal workers, inflating wages independent of bargaining power.

2. **Omitted state characteristics:** States with strong enforcement may differ in political economy, industrial structure, or historical institutions in ways that independently affect wages.

I address these concerns through (i) rich controls for observable district characteristics, (ii) robustness checks with human capital proxies, (iii) placebo tests on non-subcontracting sectors, and (iv) structural estimation that explicitly models the bargaining mechanism. While these strategies mitigate but do not eliminate endogeneity concerns, they provide the necessary empirical moments to discipline the structural model in Stage 2, where the parameters μ (bargaining power) and ρ (technological substitutability) are formally identified through equilibrium restrictions. All variables are constructed as described in Appendix A. Monetary values are deflated to constant 2011-12 prices using industry-specific WPI and CPI-IW indices.

Baseline Specification

The baseline reduced-form specification tests whether the transmission of formal sector productivity gains to informal wages is conditional on the state's regulatory environment. I estimate:

$$\ln(w_{\text{inf},dsi}) = \beta_0 + \beta_1 E_s + \beta_2 \ln(A_{F,dsi}) + \beta_3 (E_s \times \ln A_{F,dsi}) + \mathbf{X}'_{dsi}\gamma + \theta_i + \varepsilon_{dsi} \tag{4.1}$$

where:

- $w_{\text{inf},dsi}$ = Average informal wage in district d, state s, industry i (2-digit NIC)
- E_s = State-level enforcement intensity (inspector density or Besley-Burgess index)
- $A_{F,dsi}$ = Mean total factor productivity of formal firms in cell (d, s, i)

- $\mathbf{X}'_{dsi}$ = Vector of district-level controls:
  - State GDP per capita
  - Urbanization rate (share of urban population)
  - Ruling party ideology (pro-labor = 1, pro-business = 0)
- θ_i = Industry fixed effects (NIC 2-digit)
- ε_{dsi} = Error term, clustered at the state level

Note on Fixed Effects Strategy: I use **industry fixed effects only**, not district fixed effects. Including district fixed effects would perfectly absorb state-level variation in E_s, making β_1 unidentified. The trade-off is that I must rely on observable controls ($\mathbf{X}_{dsi}$) to mitigate omitted variable bias from unobserved district characteristics. In Section 4.2.1, I demonstrate robustness using state fixed effects, which identifies β_3 (the interaction) while sacrificing identification of β_1 (the main effect).

Standard Errors: All specifications cluster standard errors at the **state level** to account for within-state correlation in the error term, as E_s varies only across states.

Coefficients of Interest:

- β_1**:** Direct effect of enforcement on informal wages, holding formal productivity constant. This captures the "bargaining floor" effect—whether strong institutions raise wages even for workers supplying to low-productivity formal firms.
- β_2**:** Pass-through of formal productivity to informal wages in **low-enforcement states** (where $E_s \approx 0$). Captures baseline rent-sharing.
- β_3**: Key parameter.** Additional pass-through in high-enforcement states. A positive and significant β_3 implies that stronger labor institutions allow informal workers to capture a greater share of the surplus generated by formal-sector growth. This is the empirical signature of institutional bargaining power.

Hypothesis: $\beta_3 > 0$. In high-enforcement states, informal workers capture a larger fraction of formal productivity gains.

Interpretation: If $\beta_3 = 0$, enforcement does not alter the distribution of surplus—it merely shifts wage levels. If $\beta_3 > 0$, enforcement fundamentally changes the bargaining structure, consistent with the theoretical prediction that $\mu_s = \beta_0 + \beta_1 E_s$ (Equation 4).

Robustness Checks 1: Addressing Composition Bias

A concern is that positive pass-through may reflect **worker sorting** rather than bargaining power. If high-enforcement states attract higher-skilled workers—who would earn more regardless of institutions—then β_3 could be biased upward.

To test this, I augment the baseline specification with district-level human capital controls from the Periodic Labour Force Survey (PLFS 2023-24):

$$\begin{aligned}\ln(w_{\text{inf},dsi}) =& \beta_0 + \beta_1 E_s + \beta_2 \ln(A_{F,dsi}) + \beta_3 (E_s \times \ln A_{F,dsi}) \\ &+ \gamma_1 \text{Literacy}_d + \gamma_2 \text{SecondaryEdu}_d + \mathbf{X}'_{dsi}\phi + \theta_i + \varepsilon_{dsi}\end{aligned} \tag{4.2}$$

where:

- Literacy_d = District literacy rate (% of population aged 15+)
- SecondaryEdu_d = Share of workforce with secondary education

Caveat on Interpretation: District-level education variables (H_d) may be positively correlated with state-level enforcement (E_s), as high-enforcement states like Kerala systematically have better education systems. Adding H_d controls may **over-control** if enforcement partly works *through* improving education or if both are driven by common political economy factors. Therefore, I interpret this as a **sensitivity check** rather than a definitive decomposition.

Test: If β_3 remains positive and significant after controlling for H_d, it suggests the wage premium contains an institutional (bargaining) component beyond pure skill composition.

Standard errors: Clustered at the state level.

Robustness Check 2: Distinguishing Bargaining from Market Clearing

A second concern is that enforcement may affect wages through **demand shifts** rather than bargaining power. Specifically: strict enforcement makes formal hiring costly → for-

mal firms substitute toward informal outsourcing → increased demand bids up informal wages through standard market mechanisms.

To distinguish the **Bargaining Channel** (price effect) from the **Supply-Demand Channel** (quantity effect), I estimate the effect of enforcement on outsourcing intensity:

$$\ln(Q_{\text{out},dsi}) = \alpha_0 + \alpha_1 E_s + \alpha_2 \ln(A_{F,dsi}) + \alpha_3 (E_s \times \ln A_{F,dsi}) + \mathbf{X}'_{dsi}\phi + \theta_i + \varepsilon_{dsi} \tag{4.3}$$

where:

- $Q_{\text{out},dsi}$ = Outsourcing intensity = $\frac{\text{Cost of job work (ASI Block F)}}{\text{Total output}}$

Interpretation:

Table 4.1: Distinguishing Bargaining from Demand Channels

Channel	**Prediction**
Bargaining dominates	Enforcement raises wages but does not increase outsourcing quantity $\beta_3 > 0$ in wage equation, $\alpha_3 \approx 0$ in quantity equation
Demand shift dominates	Enforcement increases outsourcing, which bids up wages $\alpha_3 > 0$ in quantity equation

If I find $\beta_3 > 0$ (wage effect) but $\alpha_3 \approx 0$ (no quantity effect), it confirms that enforcement operates through **price determination** (bargaining power) rather than through shifts in labor demand.

Standard errors: Clustered at the state level.

Robustness Check 3: Within-State Variation (State Fixed Effects)

As an alternative identification strategy, I exploit **within-state, across-district** variation by replacing industry fixed effects with state fixed effects:

$$\ln(w_{\text{inf},dsi}) = \beta_2 \ln(A_{F,dsi}) + \beta_3 (E_s \times \ln A_{F,dsi}) + \mathbf{X}'_{dsi}\gamma + \delta_s + \theta_i + \varepsilon_{dsi} \tag{4.4}$$

where:

- δ_s = State fixed effects

Trade-off: State fixed effects absorb all time-invariant state characteristics (including E_s), so β_1 (the main effect of enforcement) is no longer identified. However, β_3 (the interaction) **is** identified because it varies across districts through $A_{F,dsi}$.

Advantage: This specification controls for unobserved state-level heterogeneity (political history, culture, industrial composition) that may confound the relationship between enforcement and wages.

Test: If β_3 remains significant with state fixed effects, it suggests the differential pass-through is not driven by omitted state characteristics.

Standard errors: Clustered at the state level.

Placebo Test: Capital-Intensive Sectors

As a falsification test, I estimate the baseline specification on the **Basic Metals sector (NIC 24)**. Unlike textiles, this sector has:

- High capital intensity
- Minimal informal subcontracting
- Vertically integrated production (tasks cannot be easily outsourced)

Since the "Adverse Incorporation" mechanism is **structurally absent** in this sector, enforcement should not affect the productivity-wage relationship:

$$\ln(w_{\text{inf},dsi}) = \beta_0 + \beta_1 E_s + \beta_2 \ln(A_{F,dsi}) + \beta_3 (E_s \times \ln A_{F,dsi}) + \mathbf{X}'_{dsi}\gamma + \theta_i + \varepsilon_{dsi} \tag{4.5}$$

for $i \in \{$**Basic Metals (NIC 24)**$\}$

Hypothesis: $\beta_3 \approx 0$ (statistically indistinguishable from zero).

Interpretation: If I find a significant β_3 in Basic Metals, it suggests my results are driven by **general state-level wage trends** (e.g., Kerala pays higher wages across all sectors) rather than the specific supply-chain bargaining channel I hypothesize for textiles.

Standard errors: Clustered at the state level.

Robustness Check 4: Testing the Job Availability Constraint

To test whether enforcement's effectiveness depends on the availability of formal sector jobs, I augment the baseline specification with state-level formal employment density:

$$\ln(w_{\text{inf},dsi}) = \beta_0 + \beta_1 E_s + \beta_2 D_s + \beta_3 (E_s \times D_s) + \beta_4 \ln(A_{F,dsi}) + \beta_5 (E_s \times \ln A_{F,dsi}) + \mathbf{X}'_{dsi}\gamma + \theta_i + \varepsilon_{dsi} \tag{4.6}$$

where:

- $D_s = \frac{\text{Formal employment in textiles (ASI)}}{\text{Working age population}}$ at state level

Interpretation:

- $\beta_1 > 0$: Enforcement raises wages (credibility effect)
- $\beta_2 > 0$: More formal jobs raise wages (availability effect)
- $\beta_3 > 0$: **Critical test** - enforcement only helps when formal jobs exist

If β_3 is positive and significant, it validates the concern that enforcement alone is insufficient without job availability. This provides empirical evidence for the theoretical mechanism discussed in Section 3.1.5.

Robustness Check 5: Testing Strategic Cost Avoidance

To test whether firms engage in strategic substitution to avoid regulatory costs, I estimate a quadratic relationship between enforcement and outsourcing:

$$\ln(Q_{\text{out},dsi}) = \alpha_0 + \alpha_1 E_s + \alpha_2 E_s^2 + \alpha_3 \text{Wage_Premium}_{dsi} + \alpha_4 \ln(A_{F,dsi}) + \mathbf{X}'_{dsi}\phi + \theta_i + \varepsilon_{dsi} \tag{4.7}$$

where:

- $\text{Wage_Premium}_{dsi} = \ln(w_{\text{formal},dsi}) - \ln(w_{\text{inf},dsi})$

Prediction under strategic cost-avoidance:

- If $\alpha_1 > 0$ and $\alpha_2 \geq 0$: Firms outsource MORE when enforcement is stricter (regulatory arbitrage dominates)
- If $\alpha_1 < 0$: Firms outsource LESS (enforcement encourages formalization)

A positive α_1 provides direct evidence for the "capitalist logic" critique: formal firms strategically increase outsourcing to circumvent labor regulations.

An alternative test for strategic cost-cutting would examine inventory volatility gaps between formal and informal firms. However, ASUSE data lacks reliable sales volatility measures for informal enterprises. The outsourcing intensity regression provides a more direct test using observed firm behavior.

All models cluster standard errors at the state level.

Robustness Check: Does μ Vary with Formal Density?

To validate whether the estimated bargaining power depends on job availability, I conduct a post-estimation analysis. After obtaining state-level bargaining power estimates $\hat{\mu}_s$ from the SMM procedure, I regress:

$$\hat{\mu}_s = \delta_0 + \delta_1 D_s + \delta_2 (E_s \times D_s) + \varepsilon_s \tag{4.8}$$

Interpretation:

- $\delta_2 > 0$: Enforcement effectiveness depends on job availability
- $\delta_2 \approx 0$: Job availability does not moderate enforcement effects

Table 4.2: Summary of Reduced-Form Specifications

Model	**LHS**	**Key RHS Variables**	**Fixed Effects**	**Sample**
(1) Baseline	$\ln(w_{\text{inf}})$	$E_s \times \ln(A_F)$	Industry	Textiles
(2) + Human Capital	$\ln(w_{\text{inf}})$	$E_s \times \ln(A_F)$, Literacy, Education	Industry	Textiles
(3) Bargaining vs. Demand	$\ln(Q_{\text{out}})$	$E_s \times \ln(A_F)$	Industry	Textiles
(4) Job Availability	$\ln(w_{\text{inf}})$	$E_s, D_s, E_s \times D_s$	Industry	Textiles
(5) Strategic Substitution	$\ln(Q_{\text{out}})$	E_s, E_s^2, Wage Premium	Industry	Textiles
(6) State FE	$\ln(w_{\text{inf}})$	$E_s \times \ln(A_F)$	State + Industry	Textiles
(7) Placebo	$\ln(w_{\text{inf}})$	$E_s \times \ln(A_F)$	Industry	Basic Metals

This test directly addresses whether the "job availability constraint" binds in practice. If δ_2 is positive and significant, it confirms that formal density moderates the relationship between enforcement and bargaining power, validating the theoretical concern while showing that μ_s captures this in a reduced-form manner.

4.2.2 Stage 2: Structural Estimation via Simulated Method of Moments (SMM)

Motivation: From Correlations to Causal Parameters

The reduced-form analysis establishes **conditional correlations** consistent with the bargaining hypothesis. However, it cannot:

1. Decompose low informal wages into “technological” (low ρ) versus “power” (low μ) components

2. Predict the effects of novel policies (e.g., a 20% increase in enforcement) due to the Lucas Critique

3. Quantify welfare trade-offs (employment loss vs. wage gain) under counterfactual regulations

To address these limitations, I structurally estimate the parameter vector:

$$\Theta = \{\rho, \gamma, \beta_0, \beta_1\} \tag{4.9}$$

where:

- ρ = Substitution parameter in the CES production function (Equation 1)
- γ = Share parameter for formal labor
- β_0, β_1 = Parameters of the bargaining power function $\mu_s = \beta_0 + \beta_1 E_s$ (Equation 5)

These "deep parameters" are **invariant to policy changes** and thus allow for counterfactual simulations.

Identification Logic: Mapping Moments to Parameters

The credibility of structural estimation rests on selecting moments that provide **unique variation** for each parameter. I follow the principle of **informal identification** (Rust 2010): each parameter should be primarily determined by a distinct feature of the data.

Note: The reduced-form coefficient β_3 does **not** directly enter Θ but serves as a **target moment** to validate the structural model. Specifically, I require that simulated data, when subjected to Equation (4.1), yields a β_3 close to the observed value.

Simulated Method of Moments (SMM) Algorithm

The SMM estimator minimizes the **weighted distance** between empirical moments (computed from ASI-ASUSE data) and simulated moments (generated by the structural model):

$$\hat{\Theta} = \arg\min_{\Theta} \left[M_{\text{data}} - M_{\text{model}}(\Theta)\right]' W \left[M_{\text{data}} - M_{\text{model}}(\Theta)\right]$$

where:

Table 4.3: Identification of Structural Parameters

Parameter	Economic Meaning	Identifying Moment
ρ	Technological substitutability between formal labor and informal outsourcing	Variance of outsourcing intensity: $\text{Var}(\ln Q_{\text{out},dsi})$
γ	Share of value-added produced in-house vs. outsourced	Mean outsourcing intensity: $\mathbb{E}[\ln Q_{\text{out},dsi}]$
β_0	Baseline bargaining power (no enforcement)	Mean formal-informal wage gap: $\mathbb{E}[\ln w_{\text{formal},dsi} - \ln w_{\text{inf},dsi}]$
β_1	Sensitivity of bargaining power to enforcement	Covariance: $\text{Cov}(E_s, \ln w_{\text{inf},dsi} \mid \ln A_{F,dsi})$

- $M_{\text{data}} = (m_1, m_2, m_3, m_4, m_5)'$ = Vector of empirical moments (see Table 4.4)
- $M_{\text{model}}(\Theta) = (\tilde{m}_1(\Theta), \ldots, \tilde{m}_5(\Theta))'$ = Vector of simulated moments for parameter guess Θ
- W = Weighting matrix (specification detailed below)

Two-Step Estimation Procedure Following Hansen (1982) and standard SMM practice, I employ a **two-step procedure** to construct the optimal weighting matrix:

Step 1: Identity Weighting (Initial Estimation) In the first stage, set the weighting matrix to the identity matrix:

$$W^{(1)} = I_{5\times 5}$$

This gives equal weight to all moments and yields the initial parameter estimate:

$$\hat{\Theta}^{(1)} = \arg\min_{\Theta} \left[M_{\text{data}} - M_{\text{model}}(\Theta)\right]' I \left[M_{\text{data}} - M_{\text{model}}(\Theta)\right]$$

Rationale: The identity matrix provides a computationally stable starting point and does not require prior knowledge of moment variances. However, it is **statistically inefficient** because it ignores the fact that different moments have different sampling variability.

Step 2: Optimal Weighting (Efficient Estimation) In the second stage, construct the **optimal weighting matrix** as the inverse of the variance-covariance matrix of the empirical moments:

$$W^* = \left[\widehat{\text{Var}}(M_{\text{data}})\right]^{-1}$$

How to estimate $\widehat{\text{Var}}(M_{\text{data}})$:

The variance-covariance matrix of the moment vector cannot be directly observed, so I estimate it via **bootstrap resampling** at the state level (to account for clustering):

1. **Resample states:** For $b = 1, \ldots, 500$:
   - Draw $S = 28$ states with replacement from the original data
   - For each resampled state, include all its districts and industries
   - This preserves the within-state correlation structure
2. **Recompute moments:** For each bootstrap sample b, compute:

$$M_{\text{data}}^{(b)} = (m_1^{(b)}, m_2^{(b)}, m_3^{(b)}, m_4^{(b)}, m_5^{(b)})'$$

3. **Estimate variance-covariance matrix:**

$$\widehat{\text{Var}}(M_{\text{data}}) = \frac{1}{B-1} \sum_{b=1}^{B} \left(M_{\text{data}}^{(b)} - \bar{M}_{\text{data}}\right) \left(M_{\text{data}}^{(b)} - \bar{M}_{\text{data}}\right)'$$

   where $\bar{M}_{\text{data}} = \frac{1}{B} \sum_{b=1}^{B} M_{\text{data}}^{(b)}$ and $B = 500$ is the number of bootstrap replications.

4. **Invert to get optimal weights:**

$$W^* = \left[\widehat{\text{Var}}(M_{\text{data}})\right]^{-1}$$

Re-estimate parameters: Using W^*, obtain the efficient estimate:

$$\hat{\Theta}^{(2)} = \arg\min_{\Theta} \left[M_{\text{data}} - M_{\text{model}}(\Theta)\right]' W^* \left[M_{\text{data}} - M_{\text{model}}(\Theta)\right]$$

Why the Optimal Weighting Matrix Matters The optimal weighting matrix W^* ensures **statistical efficiency** by down-weighting noisy moments and up-weighting precisely estimated moments. Intuitively:

- **High-variance moments** (e.g., $m_2 = \text{Var}(\ln Q_{\text{out}})$, which depends on the full distribution of outsourcing intensity) receive **low weight** because they are estimated with more sampling error
- **Low-variance moments** (e.g., $m_1 = \mathbb{E}[\ln Q_{\text{out}}]$, a simple mean) receive **high weight** because they are precisely estimated

Numerical example: Suppose after bootstrap:

$$\widehat{\text{Var}}(M_{\text{data}}) = \begin{bmatrix} 0.01 & 0.002 & \cdots \\ 0.002 & 0.50 & \cdots \\ \vdots & \vdots & \ddots \end{bmatrix}$$

This indicates m_1 (mean outsourcing) has low variance (0.01) while m_2 (variance of outsourcing) has high variance (0.50). The optimal weighting gives m_1 roughly **50 times more influence** than m_2 in the objective function.

Standard Errors for $\hat{\Theta}$ The asymptotic standard errors for the SMM estimator are given by:

$$\text{Avar}(\hat{\Theta}) = \left(G'W^*G\right)^{-1} G'W^*\widehat{\text{Var}}(M_{\text{data}})W^*G\left(G'W^*G\right)^{-1}$$

where G is the **Jacobian matrix** of moments with respect to parameters:

$$G = \left.\frac{\partial M_{\text{model}}(\Theta)}{\partial \Theta}\right|_{\Theta=\hat{\Theta}}$$

In practice: I estimate G numerically using finite differences and compute standard errors via the formula above. However, for robustness, I also report **bootstrap standard errors** (as described in Section 4.2.2, Step 6), which do not rely on asymptotic approximations.

Computational Notes **Matrix inversion stability:** If $\widehat{\text{Var}}(M_{\text{data}})$ is nearly singular (e.g., if two moments are highly correlated), direct inversion may be numerically unstable. I address this by:

1. **Regularization:** Add a small ridge term:

$$W^* = \left[\widehat{\text{Var}}(M_{\text{data}}) + \lambda I\right]^{-1}$$

   where $\lambda = 10^{-4}$ is chosen to stabilize inversion without materially affecting results.

2. **Condition number check:** Verify that $\text{condition}(\widehat{\text{Var}}(M_{\text{data}})) < 100$. If the condition number is too high, it signals that some moments are redundant and should be removed.

Convergence tolerance: The minimization algorithm terminates when:

$$\|\nabla D(\Theta)\| < 10^{-5} \quad \text{and} \quad |D(\Theta^{(t)}) - D(\Theta^{(t-1)})| < 10^{-6}$$

Table 4.4: SMM Estimation Workflow

Stage	**Weighting Matrix**	**Purpose**
Step 1	$W = I$	Obtain consistent (but inefficient) initial estimate
Bootstrap	Resample states 500×	Estimate $\widehat{\text{Var}}(M_{\text{data}})$
Step 2	$W^* = [\widehat{\text{Var}}(M_{\text{data}})]^{-1}$	Obtain efficient estimate with optimal weights
Standard Errors	Bootstrap $\hat{\Theta}$ 500×	Compute confidence intervals

Summary: Estimation Workflow **Reported results:** The main text reports $\hat{\Theta}^{(2)}$ (efficient estimates from Step 2) with bootstrap standard errors. Appendix tables show $\hat{\Theta}^{(1)}$ (Step 1 estimates) for comparison to assess sensitivity to weighting.

Simulation Procedure

For each parameter guess $\Theta = \{\rho, \gamma, \beta_0, \beta_1\}$, I generate a synthetic dataset that mirrors the structure of the ASI-ASUSE data. The simulation operates at the **district-industry level**

(d,s,i)—the same unit of observation as the empirical analysis—to ensure moment comparability.

Step 1: Initialize State-Level Institutions Draw $S = 28$ states with observed enforcement levels $\{E_s\}_{s=1}^{28}$ from the actual data. For each state s, calculate the bargaining power parameter:

$$\mu_s = \beta_0 + \beta_1 E_s$$

This endogenizes bargaining power as a function of institutional quality, consistent with the theoretical framework (Equation 5).

Step 2: Simulate Firm Decisions at the District-Industry Level For each state $s = 1,\ldots,28$:

For each district d in state s (average $D_s \approx 15$ districts per state):

For each industry $i \in \{$NIC 13 (Textiles), NIC 14 (Apparel)$\}$:

The cell (d,s,i) represents one observation in the simulated dataset, mirroring the empirical structure.

2a. Draw Formal Firms Generate $N = 100$ formal firms within cell (d,s,i) with productivity drawn from a log-normal distribution:

$$A_{F,n} \sim \text{LogNormal}(\mu_A, \sigma_A) \quad \text{for } n = 1,\ldots,100$$

Calibration: The parameters μ_A and σ_A are set to match the empirical mean and standard deviation of log-TFP in the ASI data for industry i.

2b. Observe Market Conditions For each cell (d,s,i):

- **Formal wage:** $w_F = w_F^{\min}(s,i)$ — the statutory minimum wage for industry i in state s, from labor ministry data
- **Informal marginal cost:** $MC_{\text{inf},dsi}$ — the district-industry specific cost of informal production, computed as:

$$MC_{\text{inf},dsi} = \frac{\text{Median operating expenses per unit output in ASUSE cell } (d,s,i)}{\text{Average labor productivity in ASUSE cell } (d,s,i)}$$

This approximates the cost that formal firms face when outsourcing to this district-industry's informal sector.

- **Output price:** Normalized to $P = 1$ (consistent with revenue-based TFP measurement; see Appendix A.2)

2c. Solve Each Firm's Optimization Problem For each firm $n = 1, \ldots, 100$ in cell (d,s,i):

Firm's Problem:

$$\max_{L_F, M_{\text{inf}}} \quad \pi_n = P \cdot Y_{F,n} - w_F \cdot L_{F,n} - p_{\text{sub}} \cdot M_{\text{inf},n}$$

subject to the CES production technology:

$$Y_{F,n} = A_{F,n} \left[\gamma L_{F,n}^{\rho} + (1-\gamma) M_{\text{inf},n}^{\rho} \right]^{1/\rho}$$

First-Order Conditions:

1. **For formal labor:**

$$P \cdot \frac{\partial Y_{F,n}}{\partial L_{F,n}} = w_F$$

This implicitly defines the optimal formal labor demand:

$$L_{F,n}^{*} = L_F^{*}(A_{F,n}, w_F, p_{\text{sub}}; \rho, \gamma)$$

2. **For informal outsourcing:**

$$P \cdot \frac{\partial Y_{F,n}}{\partial M_{\text{inf},n}} = p_{\text{sub}}$$

This implicitly defines the optimal outsourcing demand:

$$M^*_{\text{inf},n} = M^*_{\text{inf}}(A_{F,n}, w_F, p_{\text{sub}}; \rho, \gamma)$$

Marginal Revenue Product:

The marginal revenue product of outsourcing for firm n is:

$$MRP_{F,n} = P \cdot \frac{\partial Y_{F,n}}{\partial M_{\text{inf},n}}\bigg|_{(L^*_{F,n}, M^*_{\text{inf},n})}$$

With $P = 1$ and the CES form, this simplifies to:

$$MRP_{F,n} = A_{F,n} \cdot (1-\gamma) \cdot \left[\gamma L^{*\rho}_{F,n} + (1-\gamma) M^{*\rho}_{\text{inf},n}\right]^{(1-\rho)/\rho} \cdot M^{*(\rho-1)}_{\text{inf},n}$$

2d. Determine Bargaining Equilibrium Price The subcontracting price is determined by Nash bargaining (Equation 2):

$$p_{\text{sub},n} = (1-\mu_s) \cdot MC_{\text{inf},dsi} + \mu_s \cdot MRP_{F,n}$$

Interpretation:

- When $\mu_s = 0$ (weak enforcement), $p_{\text{sub}} = MC_{\text{inf}}$ — formal firms extract all surplus (monopsony)
- When $\mu_s = 1$ (strong enforcement), $p_{\text{sub}} = MRP_F$ — informal suppliers capture full marginal product (competitive)

Computational Implementation:

Since p_{sub} appears on both sides of the FOCs (through M^*_{inf}), this is a **fixed-point problem**. I solve it iteratively:

1. Initialize: Guess $p^{(0)}_{\text{sub}} = MC_{\text{inf},dsi}$
2. For iteration $t = 1, 2, \ldots$:

- Solve FOCs given $p_{\text{sub}}^{(t-1)} \rightarrow$ obtain $(L_F^{*(t)}, M_{\text{inf}}^{*(t)})$
- Calculate $MRP_F^{(t)}$ at these quantities
- Update: $p_{\text{sub}}^{(t)} = (1-\mu_s)MC_{\text{inf}} + \mu_s MRP_F^{(t)}$
- Stop if $|p_{\text{sub}}^{(t)} - p_{\text{sub}}^{(t-1)}| < 10^{-6}$

2e. Store Firm-Level Outcomes For each firm n, store:

- Outsourcing intensity: $Q_{\text{out},n} = \frac{M_{\text{inf},n}^*}{L_{F,n}^* + M_{\text{inf},n}^*}$
- Informal wage proxy: $w_{\text{inf},n} = p_{\text{sub},n}$ (under the assumption that subcontracting price equals effective wage paid to informal workers)

Step 3: Aggregate to District-Industry Level For each cell (d,s,i), compute:

$$Q_{\text{out},dsi} = \frac{1}{N}\sum_{n=1}^{N} Q_{\text{out},n}$$

$$w_{\text{inf},dsi} = \frac{1}{N}\sum_{n=1}^{N} w_{\text{inf},n}$$

$$A_{F,dsi} = \frac{1}{N}\sum_{n=1}^{N} A_{F,n}$$

$$w_{\text{formal},dsi} = w_F \quad \text{(constant within state-industry)}$$

This produces a synthetic dataset with the **same structure** as the empirical ASI-ASUSE panel: observations at the (d,s,i) level.

Step 4: Calculate Simulated Moments Using the aggregated synthetic data, compute:

Moment 1: Mean outsourcing intensity

$$\tilde{m}_1(\Theta) = \mathbb{E}[\ln Q_{\text{out},dsi}] = \frac{1}{N_{\text{cells}}}\sum_{d,s,i} \ln Q_{\text{out},dsi}$$

Moment 2: Variance of outsourcing intensity

$$\tilde{m}_2(\Theta) = \text{Var}(\ln Q_{\text{out},dsi})$$

Moment 3: Mean wage gap

$$\tilde{m}_3(\Theta) = \mathbb{E}[\ln w_{\text{formal},dsi} - \ln w_{\text{inf},dsi}]$$

Moment 4: Covariance of enforcement and wages (conditional on productivity)

First, residualize wages with respect to productivity:

$$\tilde{w}_{\text{inf},dsi} = \ln w_{\text{inf},dsi} - \hat{\beta} \ln A_{F,dsi}$$

where $\hat{\beta}$ is the OLS coefficient from regressing $\ln w_{\text{inf}}$ on $\ln A_F$ in the simulated data. Then:

$$\tilde{m}_4(\Theta) = \text{Cov}(E_s, \tilde{w}_{\text{inf},dsi})$$

Moment 5: Reduced-form interaction coefficient

Run the baseline reduced-form regression on the simulated data:

$$\ln(w_{\text{inf},dsi}) = b_0 + b_1 E_s + b_2 \ln(A_{F,dsi}) + b_3(E_s \times \ln A_{F,dsi}) + \theta_i + \varepsilon_{dsi}$$

Extract the coefficient:

$$\tilde{m}_5(\Theta) = \hat{b}_3$$

Step 5: Compute Objective Function The SMM objective is:

$$D(\Theta) = [M_{\text{data}} - M_{\text{model}}(\Theta)]' W [M_{\text{data}} - M_{\text{model}}(\Theta)]$$

where $M_{\text{data}} = (m_1, m_2, m_3, m_4, m_5)'$ from ASI-ASUSE and $M_{\text{model}}(\Theta) = (\tilde{m}_1, \tilde{m}_2, \tilde{m}_3, \tilde{m}_4, \tilde{m}_5)'$ from the simulation.

Step 6: Optimization and Standard Errors

1. **Numerical minimization:** Use the Nelder-Mead simplex algorithm to find:

$$\hat{\Theta} = \arg\min_{\Theta} D(\Theta)$$

   Given the computational intensity (each function evaluation requires simulating ∼10,000 firms), I impose the following constraints to aid convergence:

   - $\rho \in [-2, 0.99]$ (ensures $\sigma = 1/(1-\rho) > 0$)
   - $\gamma \in [0.1, 0.9]$ (both inputs are essential)
   - $\beta_0 \in [0,1]$, $\beta_1 \in [0,5]$ (bargaining power is bounded)

2. **Repeat simulations:** For each parameter guess, I generate $S = 10,000$ simulation draws (resampling firm productivities) and average the moments to reduce simulation noise.

3. **Bootstrap standard errors:**

   - Resample **states** with replacement (block bootstrap to preserve within-state correlation)
   - For each bootstrap sample $b = 1, \ldots, 500$:
     - Re-estimate $\hat{\Theta}_b$
   - Compute standard errors as: $SE(\hat{\Theta}) = \sqrt{\text{Var}(\{\hat{\Theta}_1, \ldots, \hat{\Theta}_{500}\})}$

4.2.3 Model Validation and Fit Diagnostics

The credibility of any structural model depends on its ability to replicate the empirical patterns it seeks to explain. I assess model fit through four complementary diagnostics: moment matching precision, distributional comparison, over-identification tests, and parameter stability checks.

Moment Matching Precision

The SMM estimator minimizes the distance between empirical and simulated moments, but the **magnitude of the residual distance** at $\hat{\Theta}$ reveals whether the model can adequately explain the data. For each of the five target moments, I compute the percentage deviation:

$$\text{Deviation}_j = \frac{|m_{j,\text{data}} - m_{j,\text{model}}(\hat{\Theta})|}{|m_{j,\text{data}}|} \times 100\%$$

Benchmark: Well-specified models in the SMM literature typically achieve deviations below 10% for key moments (Gourieroux and Monfort 1996). Deviations exceeding 25% suggest potential misspecification.

Reporting: I present these deviations in a table alongside the raw moments:

Table 4.5: Moment Matching Results

Moment	**Data Value**	**Model Value**	**Deviation (%)**
m_1: Mean $\ln Q_{\text{out}}$	[Value]	[Value]	[%]
m_2: Var $\ln Q_{\text{out}}$	[Value]	[Value]	[%]
m_3: Mean $\ln(w_{\text{formal}}/w_{\text{inf}})$	[Value]	[Value]	[%]
m_4: $\text{Cov}(E_s, \ln w_{\text{inf}})$	[Value]	[Value]	[%]
m_5: Reduced-form $\hat{\beta}_3$	[Value]	[Value]	[%]

Diagnostic interpretation:

- If m_1, m_2 (quantity moments) fit well but m_3, m_4, m_5 (price moments) fit poorly $\rightarrow$ Technology parameters (ρ, γ) are well-identified, but bargaining parameters (β_0, β_1) may be misspecified
- If m_5 (the reduced-form β_3) deviates substantially $\rightarrow$ The structural model's institutional mechanism does not replicate the pass-through elasticity observed in the data

Distributional Comparison: Beyond Moments

While SMM matches **moments** of the data, it does not guarantee that the **full distribution** of outcomes is replicated. For example, the model might match the mean and variance

of informal wages but fail to capture skewness or bimodality. To assess this, I compare the empirical and simulated distributions visually and statistically.

Visual Diagnostics:

I generate kernel density plots overlaying the actual and simulated distributions for:

1. **Informal wages:** $\ln(w_{\text{inf},dsi})$
2. **Outsourcing intensity:** $\ln(Q_{\text{out},dsi})$
3. **Formal-informal wage gap:** $\ln(w_{\text{formal},dsi}) - \ln(w_{\text{inf},dsi})$

Example interpretation:

- If the simulated distribution is **too tight** (lower variance than data) $\rightarrow$ The model under-predicts the heterogeneity in bargaining outcomes, possibly because $\mu_s = \beta_0 + \beta_1 E_s$ is too simplistic (e.g., missing firm-level bargaining power)
- If the simulated distribution has the **wrong shape** (e.g., unimodal when data is bimodal) $\rightarrow$ Suggests missing heterogeneity in technology (ρ may vary across firm types)

Quantitative Test: Kolmogorov-Smirnov (KS) Statistic

To formalize the comparison, I compute the two-sample KS test statistic:

$$D_{KS} = \sup_x |F_{\text{data}}(x) - F_{\text{model}}(x)|$$

where F_{data} and F_{model} are the empirical cumulative distribution functions. Under the null hypothesis that the model-generated data comes from the same distribution as the actual data:

- $p > 0.10 \rightarrow$ Cannot reject; distributions are similar
- $p < 0.05 \rightarrow$ Reject; model fails to replicate the distribution

Reporting: I present KS statistics and p-values for each key variable:

Table 4.6: Kolmogorov-Smirnov Tests of Distributional Fit

Variable	KS Statistic	p-value	Interpretation
$\ln w_{\text{inf}}$	[Value]	[Value]	[Pass/Fail]
$\ln Q_{\text{out}}$	[Value]	[Value]	[Pass/Fail]
Wage gap ratio	[Value]	[Value]	[Pass/Fail]

Over-Identification Test: Predicting Untargeted Moments

A powerful test of model validity is to check whether it correctly predicts **moments that were not used in estimation**. This is analogous to out-of-sample validation in machine learning.

Implementation:

1. **Exclude moment** m_5 (the reduced-form β_3 coefficient) from the SMM estimation. Re-estimate Θ using only m_1, m_2, m_3, m_4.

2. **Generate predictions:** Using the restricted estimate $\hat{\Theta}_{\text{restricted}}$, simulate data and compute the model-implied β_3.

3. **Compare:** Is the model-implied β_3 close to the actual reduced-form estimate?

Formal Test: Hansen's J-Statistic

When the number of moments ($K = 5$) exceeds the number of parameters ($|\Theta| = 4$), the model is **over-identified**. Hansen's J-test evaluates whether the over-identifying restrictions are satisfied:

$$J = N \cdot \left[M_{\text{data}} - M_{\text{model}}(\hat{\Theta})\right]' \hat{W} \left[M_{\text{data}} - M_{\text{model}}(\hat{\Theta})\right]$$

where N is the effective sample size (number of district-industry cells) and $\hat{W}$ is the optimal weighting matrix. Under the null hypothesis that the model is correctly specified:

$$J \sim \chi^2(K - |\Theta|) = \chi^2(1)$$

Decision rule:

- If $p(J) > 0.10 \rightarrow$ Cannot reject; model restrictions are satisfied
- If $p(J) < 0.05 \rightarrow$ Reject; model is misspecified (some moments are incompatible with the structure)

Interpretation if J-test rejects:

- The model **cannot simultaneously** fit all five moments, suggesting either:
  - The linear bargaining function $\mu_s = \beta_0 + \beta_1 E_s$ is too restrictive
  - The CES technology is mis-specified (e.g., ρ varies across firm size)
  - Measurement error in enforcement E_s biases the estimates

Parameter Stability and Sensitivity Analysis

To assess whether the estimated parameters are robust, I conduct two checks:

A. Subsample Stability

I re-estimate Θ on subsamples:

1. **High-enforcement states only** (E_s above median)
2. **Low-enforcement states only** (E_s below median)

Test: Do the technology parameters (ρ, γ) remain stable across subsamples?

- **Yes** $\rightarrow$ Technology is truly a "deep parameter" (invariant to institutions)
- **No** $\rightarrow$ Suggests the CES assumption is violated (e.g., production technology changes with enforcement environment)

B. Sensitivity to Moment Weighting

The SMM objective function depends on the weighting matrix W. I check robustness by re-estimating with:

1. **Equal weighting:** $W = I$ (identity matrix)
2. **Diagonal weighting:** $W = \text{diag}(1/\sigma_j^2)$ where $\sigma_j^2 = \text{Var}(m_j)$

If $\hat{\Theta}$ changes substantially (e.g., $\hat{\rho}$ shifts from 0.6 to -0.2), it indicates **weak identification**—the moments do not tightly pin down the parameters.

Summary: Model Validation Checklist

The structural model passes validation if:

Table 4.7: Model Validation Checklist

Diagnostic	Pass Criterion	Implication if Failed
Moment deviations	All $< 25\%$	Poor overall fit; re-specify model
KS test (wage distribution)	$p > 0.10$	Model misses distributional features of wages
KS test (quantity distribution)	$p > 0.10$	Technology (ρ, γ) misspecified
Hansen's J-test	$p > 0.10$	Over-identifying restrictions rejected
Subsample stability	$\|\Delta\rho\| < 0.3$	Technology not invariant to institutions
Internal consistency	Simulated β_3 matches observed β_3 (within 20	Model's institutional mechanism doesn't replicate reduced-form pass-through

Reporting in the dissertation: I present:

1. **Table of moment matching** (Section 4.2.5.1)
2. **Kernel density plots** (Section 4.2.5.2)
3. **J-test result** (Section 4.2.5.3)
4. **Subsample estimates** (Section 4.2.5.4)

If the model fails any diagnostic, I discuss potential explanations and how future iterations could address them (e.g., allowing for non-linear $\mu(E_s)$ or heterogeneous ρ by firm size).

4.2.4 Policy Counterfactuals

The mathematical procedure for simulating policy shocks is detailed below; the specific economic rationale for these scenarios and their predicted impacts on the Indian textile value

chain are discussed in Section 5. The structural parameters $\hat{\Theta}$ allow me to simulate the effects of policy changes that have **never been observed** in the data. I consider two scenarios:

Counterfactual 1: The "Deregulation" Shock

Discuss the "Race to the Bottom" theory. If India reduces inspections to compete with Bangladesh, does the extraction rate (μ) drop so low that it offsets any gains from increased outsourcing?

Scenario: Reduce enforcement by 20% uniformly across all states:

$$E_s^{\text{new}} = 0.8 \times E_s^{\text{baseline}}$$

Mechanism:

$$\mu_s^{\text{new}} = \beta_0 + \beta_1(0.8 \times E_s) < \mu_s^{\text{baseline}}$$

Lower bargaining power shifts surplus toward formal firms.

Simulation:

1. Re-run the SMM algorithm with E_s^{new}, holding Θ fixed
2. Measure:
   - Δw_{inf} = Change in informal wages
   - ΔQ_{out} = Change in outsourcing quantity (employment effect)
   - $\Delta \pi_F$ = Change in formal firm profits

Welfare Calculation:

$$W = \underbrace{\sum_d N_{\text{inf},d} \cdot w_{\text{inf},d}}_{\text{Informal worker surplus}} + \underbrace{\sum_d \pi_{F,d}}_{\text{Formal firm profits}}$$

I report changes in W and its distribution across groups.

Counterfactual 2: The "Fair Price Floor"

Discuss the "Displacement" risk. If you force formal firms to pay more to informal workers, do they stop outsourcing and just buy more machines? (This links back to ρ parameter).

Scenario: Mandate a minimum subcontracting price 20% above marginal cost:

$$p_{\text{sub}}^{\min} = 1.2 \times MC_{\text{inf}}$$

Mechanism: If p_{sub} (from bargaining) falls below $p_{\text{sub}}^{\min}$, the price floor binds. Formal firms face higher input costs and may:

- **Substitute toward formal labor** ($L_F \uparrow$, $M_{\text{inf}} \downarrow$) if σ is high (tasks are substitutes)
- **Absorb the cost** if σ is low (tasks are complements)

Simulation:

1. Impose the price floor: $p_{\text{sub}} = \max\{(1-\mu_s)MC_{\text{inf}} + \mu_s \text{MRP}_F, 1.2MC_{\text{inf}}\}$
2. Re-solve each firm's optimization problem
3. Measure:
    - ΔM_{inf} = Change in informal employment
    - $\Delta(w_{\text{inf}} \mid \text{employed})$ = Wage gain for workers who keep jobs
    - Net welfare = Employment loss vs. income gain

Key Insight: The welfare implications depend critically on $\hat{\rho}$ (equivalently, $\hat{\sigma}$):

- If $\sigma > 3$ (high substitutability), price floors **destroy jobs**
- If $\sigma < 1.5$ (low substitutability), price floors **transfer surplus** without large employment losses

4.2.5 Identification Strategy and Validity

Endogeneity of Enforcement

A primary concern is that enforcement (E_s) is not randomly assigned. States with strong labor institutions (e.g., Kerala) differ systematically from others in terms of political ideology, historical union density, and industrial composition. These unobserved factors could independently drive higher wages, leading to omitted variable bias.While instrumental variable strategies (e.g., political discontinuities, historical labor law shocks) would strengthen causal claims, such instruments are unavailable in the ASI-ASUSE data. I therefore rely on a triangulation approach: (i) rich controls, (ii) state fixed effects, (iii) placebo tests in non-subcontracting sectors, and (iv) the SMM's over-identification structure. While this does not fully eliminate endogeneity, it provides conditional correlation estimates that are consistent with the bargaining hypothesis and inform the structural model.

Statistical Power Analysis

Given that enforcement varies at the state level ($n = 28$), the power to detect the main effect (β_1) is limited. Monte Carlo simulations suggest approximately 65% power to detect a medium effect size $\beta_1 \geq 0.15$ at the 5% level, assuming typical within-state variance. This implies a non-trivial risk of Type II error (failing to reject when the effect exists).

However, the interaction coefficient β_3—the primary parameter of interest—exploits district-level variation ($n \approx 700$ district–industry cells) through $A_{F,dsi}$ and achieves $> 90\%$ power to detect meaningful pass-through effects ($\beta_3 \geq 0.10$). Therefore, I focus interpretation on β_3 rather than β_1, consistent with the research question about institutional moderation of productivity pass-through.

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Appendix A

Appendix A: Data Appendix

A.1 Variable Construction and Measurement

This appendix details the construction of key variables and the data cleaning protocol used to harmonize the Annual Survey of Industries (ASI) and the Annual Survey of Unincorporated Sector Enterprises (ASUSE) for the 2023-24 period.

A.1.1 Variable Definitions and Deflation

All monetary values are deflated to constant 2011-12 prices to ensure that cross-sectional productivity differences are not confounded by regional price variations.

- **Output and Inputs:** Deflated using the industry-specific Wholesale Price Index (WPI).
- **Wages:** Deflated using the Consumer Price Index for Industrial Workers (CPI-IW) at the state level.

Table A.1: Mapping Theoretical Parameters to Empirical Variables

Concept	Symbol	ASI/ASUSE Variable	Measurement Goal
Productivity	A_F	Revenue-based TFP (TFPR)	Efficiency of formal "surplus"
Informal Wage	w_{inf}	Total emoluments / Hired workers (ASUSE)	The bargaining outcome
Enforcement	E_s	Inspectors per 1,000 factories	Institutional "policing power"
Outsourcing	Q_{out}	(Cost of Job Work) / (Total Inputs)	Integration of value chain

A.1.2 TFP Calculation and Parameterization

Formal sector productivity (A_F) is estimated as the Solow Residual:

$$\ln A_{F,dsi} = \ln Y_{dsi} - \theta_i \ln L_{dsi} - (1-\theta_i) \ln K_{dsi}$$

where Y is real GVA, L is total mandays, and K is net fixed assets. The labor share θ_i is calculated as the median ratio of total emoluments to GVA within each 2-digit NIC industry.

Note: This θ_i (labor's share of output) is distinct from the returns-to-scale parameter α in Equation (1).

A.1.3 Data Cleaning and Outlier Rules

To ensure the structural model is disciplined by reliable moments, the following "cleaning" filters are applied:

1. **Subcontracting Filter:** In ASUSE, we retain only enterprises that explicitly report receiving "job work" from other units. This ensures w_{inf} reflects the supply-chain wage rather than independent retail earnings.

2. **Zero-Value Filter:** Observations with zero or negative GVA, zero total assets, or zero employees are dropped, as they typically represent defunct units or reporting errors.

3. **Cell-Size Constraint:** District-industry cells (d,i) with fewer than 10 observations in either the formal (ASI) or informal (ASUSE) sample are excluded. This prevents the SMM moments from being driven by idiosyncratic "outlier" firms in small districts.

4. **Winsorization:** To mitigate the impact of extreme values (e.g., a firm reporting 10,000% productivity), all wage and productivity variables are winsorized at the 1st and 99th percentiles.

A.1.4 Moment Selection for SMM

The SMM algorithm utilizes five primary moments from this cleaned dataset:

- m_1: Mean of $\ln Q_{\text{out}}$ (identifies γ).
- m_2: Variance of $\ln Q_{\text{out}}$ (identifies substitution ρ).
- m_3: Average $\ln(w_{\text{formal}}/w_{\text{inf}})$ (identifies baseline power β_0).
- m_4: Correlation between E_s and w_{inf} (identifies β_1).
- m_5: The reduced-form interaction $\hat{\beta}_3$ from Equation 6 (Validation moment).

A.1.5 Robustness: Alternative Measures

For sensitivity analysis, I construct alternative measures:

- **Alternative Enforcement:** Inspection frequency (inspections per registered factory per year)
- **Alternative Productivity:** Gross Value Added per worker
- **Alternative Wage Measure:** Value added per worker in informal enterprises

A.2 Conclusion of Appendix A

Summary of data appendix content.